LAB 2 Submission Requirements

Provide clear names to all your figures and brief explanation wherever required. Keep your report concise within 10-15 pages.

Total: 50 points

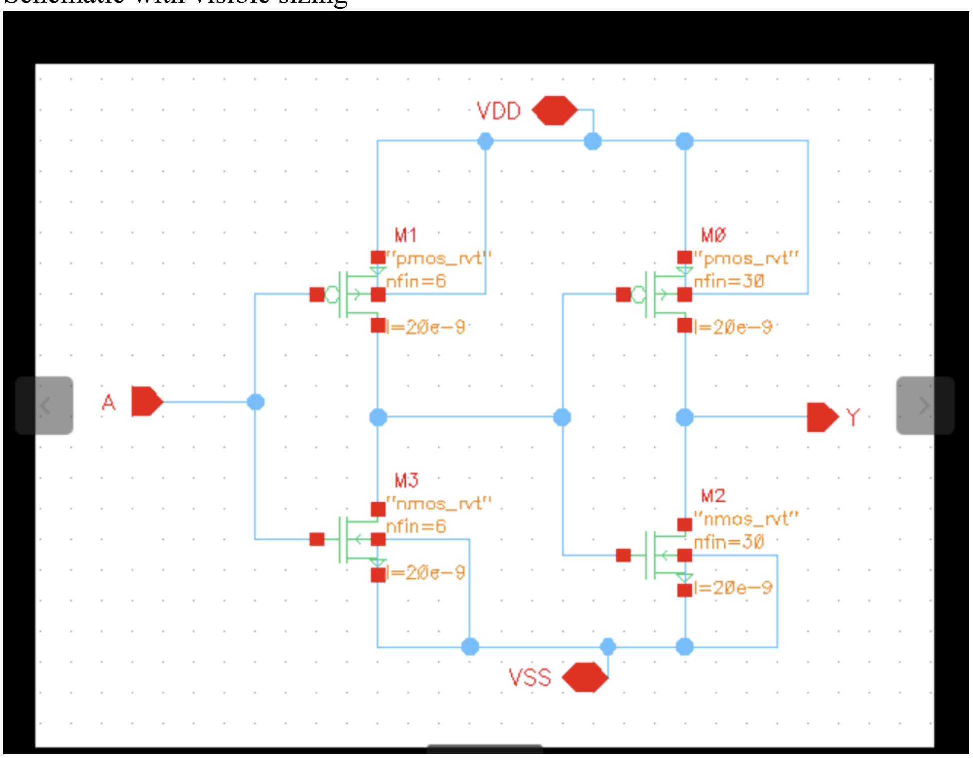
Cell Name:BUFx10 ASAP7 75t R

1	Student	Cellname	Fins at input	NMOS fins at input	PMOS fins at input	NMOS stack depth	PMOS stack depth	NMOS first stage drive	PMOS first stage drive	Fanup Between Stages	Output driv (vs. 1x inverter)
22	Jagtap, Anjali Ajay	BUFx10_ASAP7_75t_R	30	6	6	1	1	2.00	2.00	5	1

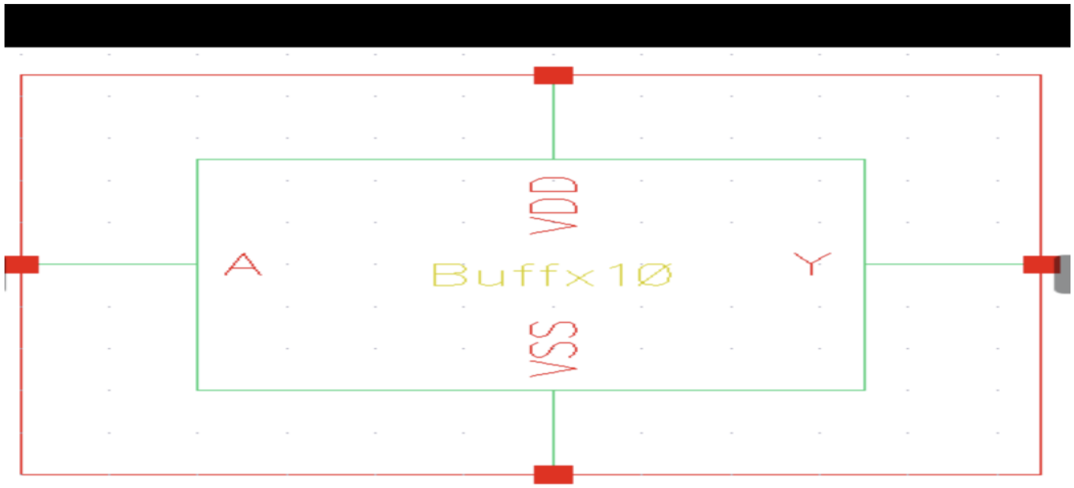
1. Schematic:
   a. Truth Table

A	AN	Y
0	1	0
1	0	1

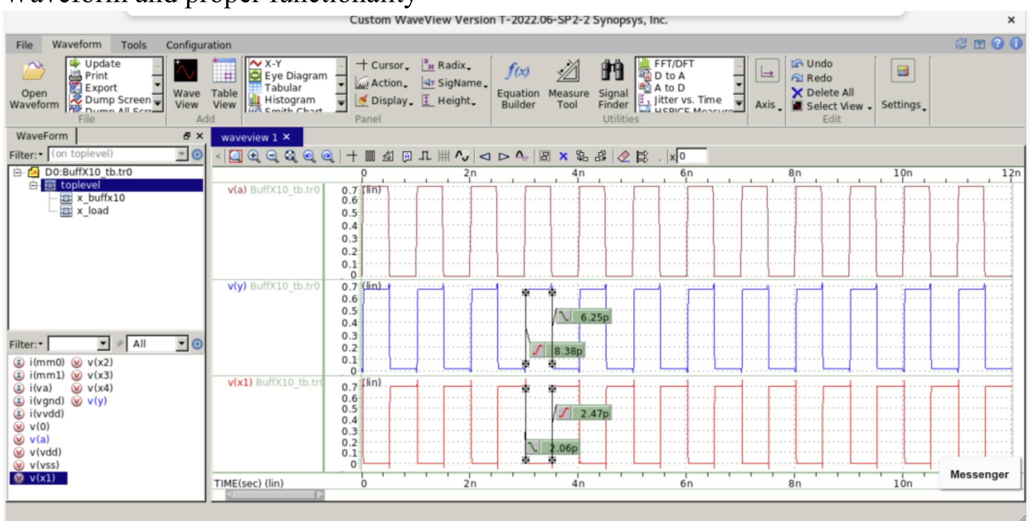
   b. Schematic with visible sizing



   c. Symbol

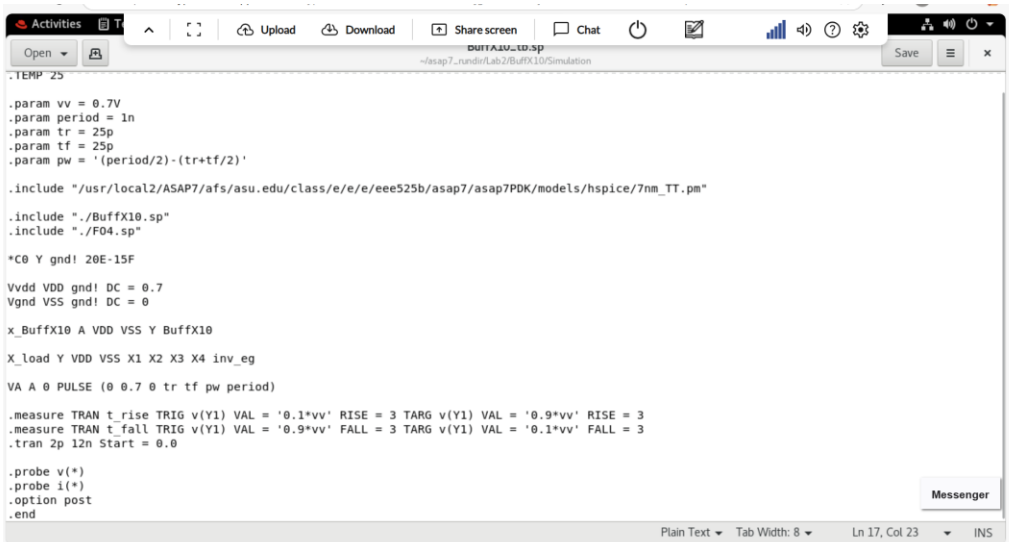


d. Waveform and proper functionality



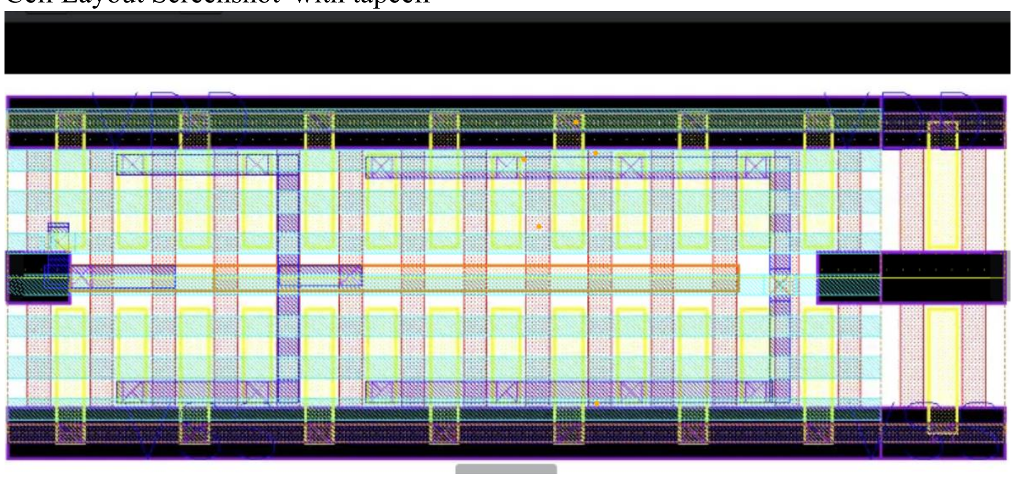
1. My cell consists of a two-stage buffer. The first stage includes 6 fins for both PMOS and NMOS, while the second stage has 30 fins.
2. Since the circuit has two inverter stages, the output is expected to be a square wave when the input is a square wave.
3. Based on the graph, the desired functionality has been achieved.
4. The input provided is a square wave, and the output obtained is also a square wave, confirming the proper operation of the circuit.

e. Testbench schematic with load

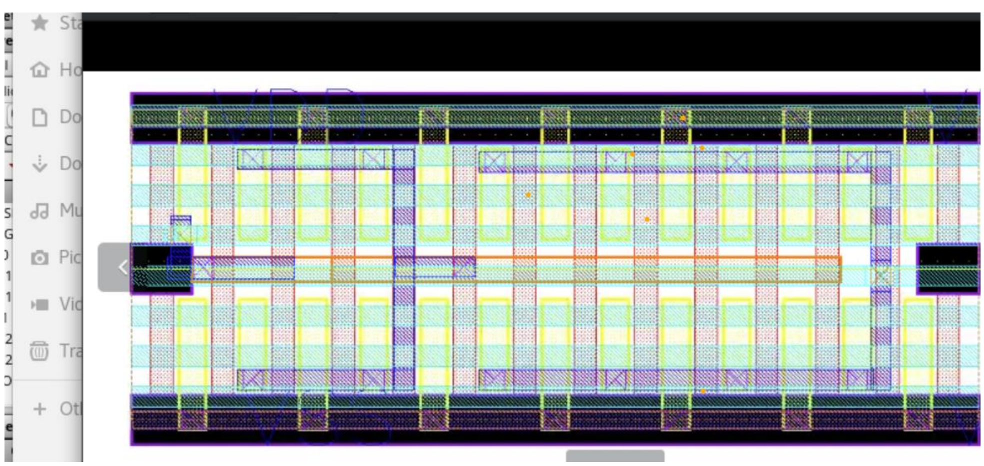


```
.TEMP 25

.param vv = 0.7V
.param period = 1n
.param tr = 25p
.param tf = 25p
.param pw = '(period/2)-(tr+tf/2)'

.include "/usr/local2/ASAP7/afs/asu.edu/class/e/e/e/eee525b/asap7/asap7PDK/models/hspice/7nm_TT.pm"

.include "./BuffX10.sp"
.include "./FO4.sp"

*C0 Y gnd! 20E-15F

Vvdd VDD gnd! DC = 0.7
Vgnd VSS gnd! DC = 0

x_BuffX10 A VDD VSS Y BuffX10

X_load Y VDD VSS X1 X2 X3 X4 inv_eg

VA A 0 PULSE (0 0.7 0 tr tf pw period)

.measure TRAN t_rise TRIG v(Y1) VAL = '0.1*vv' RISE = 3 TARG v(Y1) VAL = '0.9*vv' RISE = 3
.measure TRAN t_fall TRIG v(Y1) VAL = '0.9*vv' FALL = 3 TARG v(Y1) VAL = '0.1*vv' FALL = 3
.tran 2p 12n Start = 0.0

.probe v(*)
.probe i(*)
.option post
.end
```

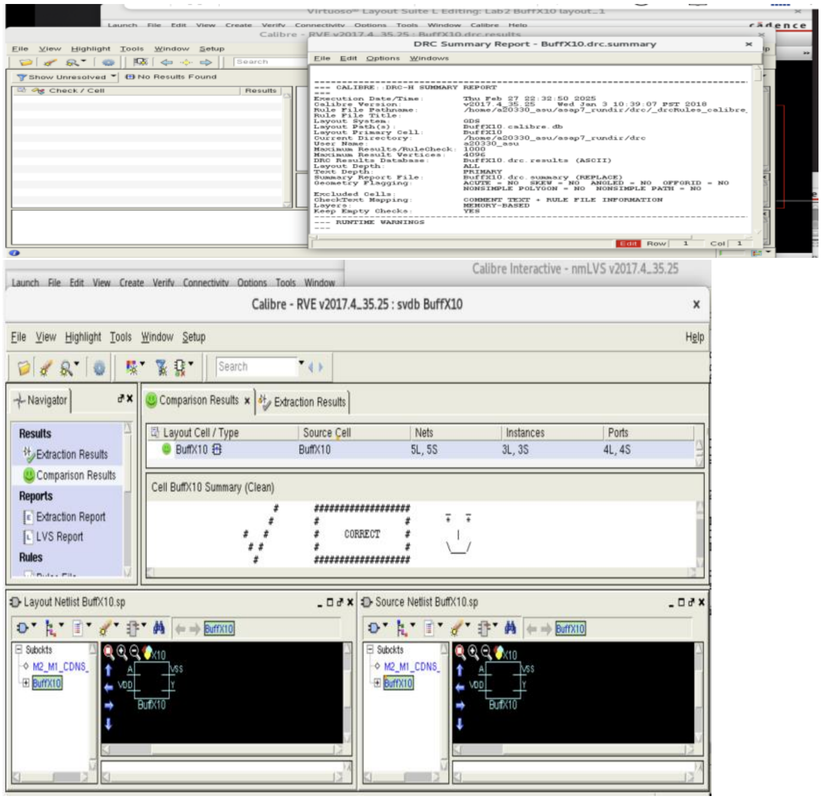
2. Layout:
   a. Cell Layout Screenshot with tapcell



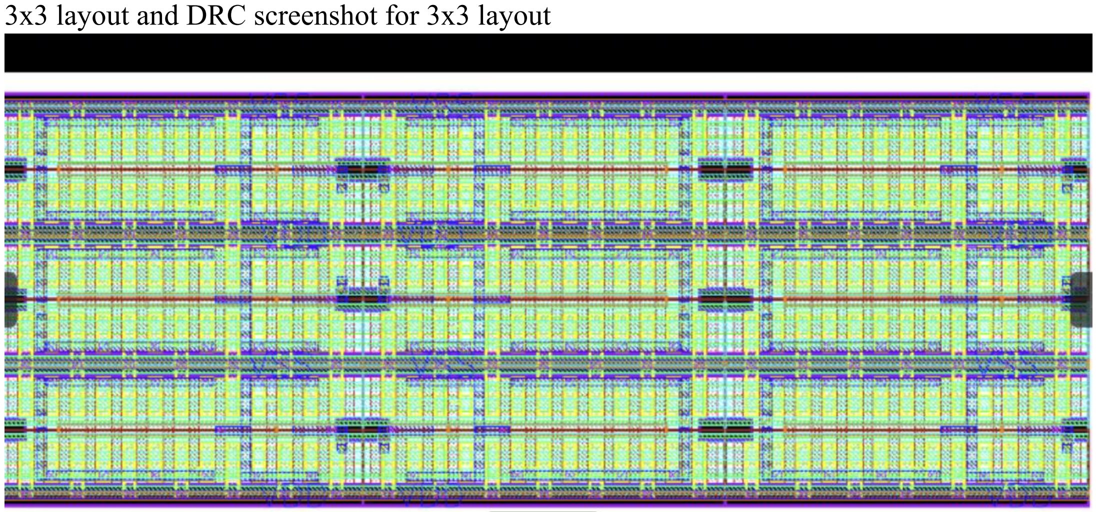
Cell Layout Screenshot without tapcell

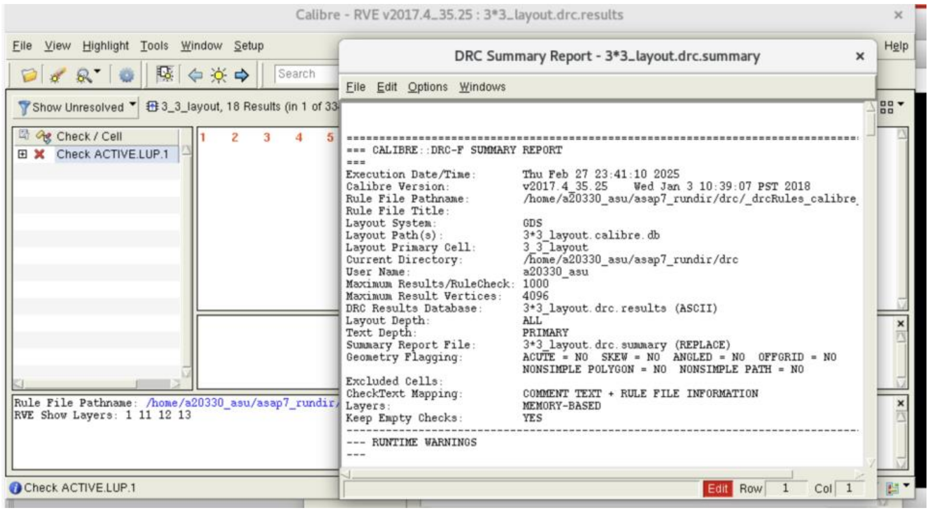


   b. DRC and LVS screenshot



c. 3x3 layout and DRC screenshot for 3x3 layout

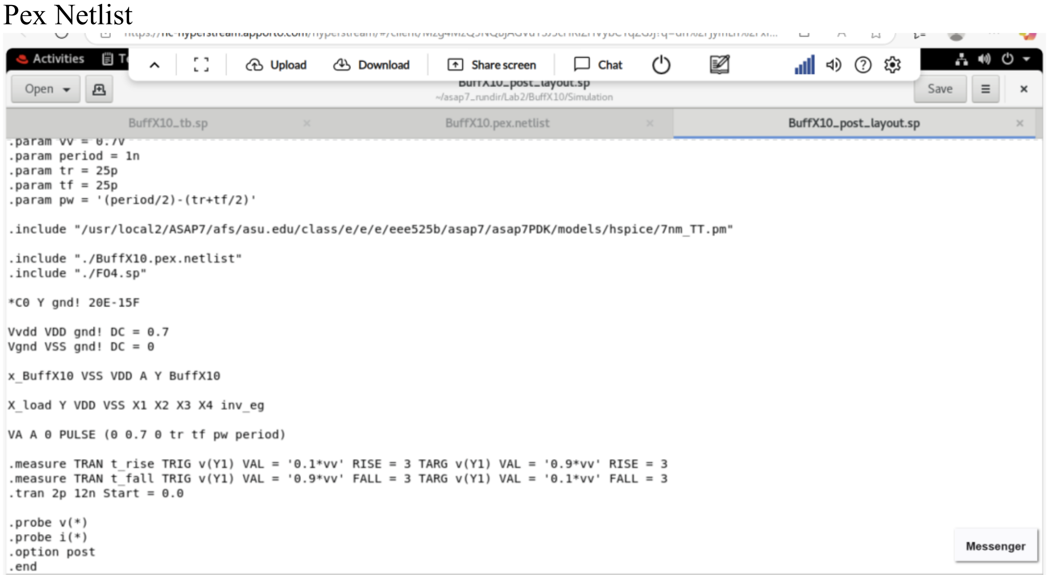




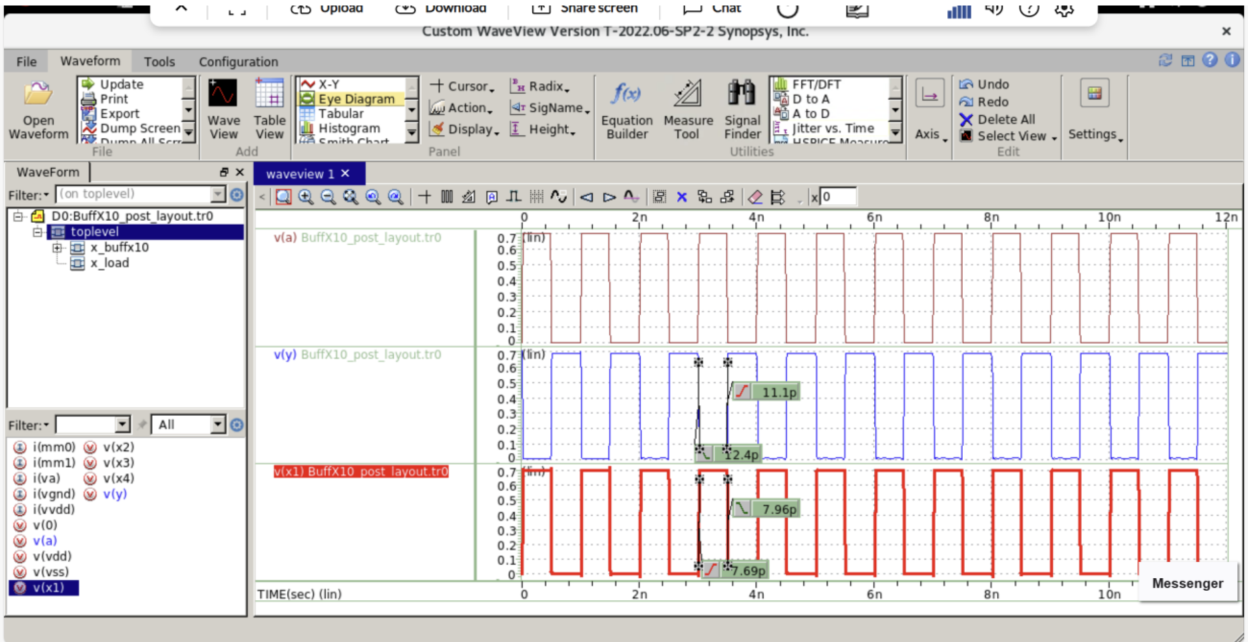
d. Full path to your design
/home/a20330_asu/asap7_rundir/Lab2/BuffX10

3. Post Layout:

a. Pex Netlist



b. Waveform and proper functionality with brief explanation



- After conducting the post-layout simulation, it was observed that the output's rise and fall times increased slightly.
- Likewise, the 50% delay between the input and output also showed a minor increase.
- This occurs because the post-layout simulation is performed using a parasitic-extracted netlist.
- Following the layout design, parasitic resistances and capacitances may arise due to metal layers, interconnects, and other factors.
- These parasitic introduce delays in signal propagation, leading to timing degradation.
- However, in pre-layout simulation, parasitic are not considered.
- As a result, delays are lower in pre-layout simulations compared to post-layout simulations.
- The variation in delay values depends on the layout design, with the smallest delays observed in an optimized layout.